

PRANAV REDDY

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SUMMARY

High-achieving Grade 12 student with strong foundations in Mathematics, Physics, and Computer Science, with experience in Python, Arduino robotics, and geolocation-based application development. Conducted independent research on gravitational-wave detection using LIGO/Virgo strain data, applying signal processing, matched filtering, and statistical analysis. Proven leadership as Head Boy and House Captain, alongside hands-on experience developing an animal rescue application with Blue Cross.

EDUCATION

Grade 12, IGCSE Program

Graduating Apr '27

Gateway The Complete School, Chennai, Tamil Nadu

AS levels: Mathematics (B), Physics (B), Chemistry (D), Computer Science (C)

IGCSE: Mathematics (A+), Physics (A+), Chemistry (A+), Computer Science (A)

SAT: 1420

CERTIFICATIONS

Ahmedabad Grand Challenges Program

May'26 - June'26

Ahmedabad University

- Explored real-world urban challenges through interdisciplinary STEM-based problem solving
- Collaborated with peers to develop innovative solutions addressing community and environmental challenges
- Engaged with experts and stakeholders to understand local issues and evaluate potential solutions
- Part of 30 students selected from over 600 students who applied pan India.

CS50 – Introduction to Computer Science

Nov '25 – Dec '25

Harvard University

- Studied core computer science concepts including algorithms, data structures, and problem-solving through C, Python, and SQL
- Completed hands-on programming assignments and problem sets under Harvard's rigorous online curriculum

Young Tech Scholars (STEM & Analytics)

May '25 – Jun '25

Plaksha University

- Engaged in an intensive STEM and analytics program covering data-driven problem solving and applied technology
- Collaborated with peers on analytics-based projects under guidance from Plaksha University faculty
- Built a weather glider under the mentorship of professors and help from peers.

Junior MBA (Business & Entrepreneurship)

Aug '23 – Oct '23

Clever Harvey

- Learned foundational business concepts including marketing, finance, and entrepreneurship strategy
- Developed a business plan and pitch as part of a hands-on entrepreneurship simulation

Robotics and Engineering Design

May '25

Texoham

- Designed and built robotics prototypes applying core engineering design principles
- Applied hands-on problem-solving skills to real-world engineering challenges during the program

TECHNICAL SKILLS

Programming: Python (Proficient), C++, Java, HTML/CSS

Software/Tools: Arduino (Robotics), Geolocation APIs

EXPERIENCE

Gateway The Complete School: Head Boy

2026 – 2027

- Elected to lead 200+ students; organized inter-school STEM competitions and sports meets
- Developed conflict resolution skills and managed large-scale logistics for school events

Gateway The Complete School: House Captain

2025 – 2026

- Led house activities and mentored peers across academic and extracurricular events
- Coordinated with faculty to represent the house in competitions and school assemblies

Blue Cross, Chennai: Animal Rescue Volunteer

2024 – Present

- Assisted in rehabilitation of distressed animals and coordinated with veterinary staff

RESEARCH & ACADEMIC PROJECTS

Research Article: “Detection of Gravitational Waves using matched filtering: An analysis of LIGO/Virgo Strain data for GW150914 and GW170817”| Published in The Curieux Review

2025 - Present

- Analysed LIGO/Virgo data from the gravitational-wave events GW150914 and GW170817.
- Used Python for data processing, filtering, matched filtering, and signal analysis.
- Used ROC curves and a confusion matrix to evaluate the accuracy of the detection method.
- Studied the limitations of the method and suggested ways to improve the detection process.

Animal Rescue & Recovery App

2025 – Present.

- Designed a mobile application to streamline injured animal reporting pitched to Blue Cross (Lead Developer)
- Implemented Geolocation APIs to provide real-time coordinates for rescue teams
- Built backend logic and database using Python/Java for shelter capacity tracking
- Reduced manual reporting time and improved rescue response coordination during pilot testing

Robotics & AI Explorations

2024 – Present

- Explored applications of AI and robotics for social good using Arduino hardware and Python software.
- Prototyped sensor-driven systems integrating microcontrollers with real-world data inputs
- Applied problem-solving skills to translate community needs into working technical solutions

ACTIVITIES

Music – Trinity College London (Keyboard)

Ongoing

- Grade 6 Keyboard certification; mastered complex rhythmic patterns and advanced music theory.
- Developed discipline and creative expression through years of structured practice, performing in school events and graded examinations

Shotokan Karate Dojo, Chennai: Black Belt Trainee & Assistant

2017 – 2025

- Completed 8+ years of rigorous training culminating in Black Belt (Shodan) certification

Football - Part of school football team

Ongoing

- Part of the school football team as a defender and represented school in multiple tournaments

Cricket - Captained club team

2021-2022

- Captained cricket club team in multiple tournaments

LANGUAGES

English (Fluent), Telugu (Fluent), Tamil (Intermediate), Hindi (Intermediate)